Day 17 Report: Multi-Agent Systems and A2A

A research team of four agents (LangGraph), an A2A Researcher, a single vs multi comparison, and a live AI Agent Office

1. Goal

Until Day 16 one agent did everything. Today the work is split across a **team of agents**, each with one job, an **orchestrator** that hands out the work, and the **A2A protocol** so agents can talk to each other across services. Then I measured whether the team is actually better than one agent.

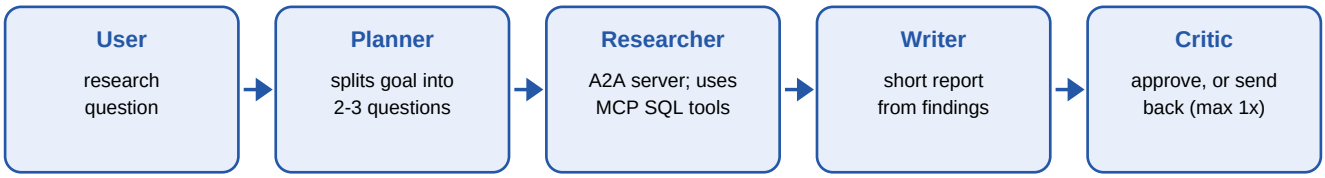
2. Theory in short

Idea	Simple meaning
Multi-agent system	Several LLM agents, each with its own role prompt, tools and permissions, working on one goal
Orchestrator-worker	One agent (Planner) splits the goal into sub-tasks and hands them to workers; it never does the work itself
Critic / reviewer loop	A separate agent checks the result and can send it back for one revision
LangGraph	Agents are nodes in a graph; edges say who runs next; shared state carries the work between them
A2A (Agent2Agent)	Open protocol for agent-to-agent calls: an **Agent Card** says who the agent is and what it can do; messages and tasks travel as JSON-RPC over HTTP

	MCP (Day 16)	A2A (Day 17)
Connects	an agent to **tools and data**	an agent to **another agent**
Discovery	tools/list	Agent Card at /.well-known/agent-card.json
Unit of work	one tool call	a task (can take many steps)
In my project	Researcher reads the customer DB	Planner delegates questions to the Researcher service

3. Architecture

Orchestrator-worker team (LangGraph)



Only the Researcher has tools. It runs inside my Day 15 harness (step budget, "tool output is data" rule, approval policy *deny*) and gets the Day 16 MCP tools through the bridge (read-only SQLite, SELECT only). Model: **openai/gpt-oss-120b on Groq** (free tier, limited to 25 calls per minute).

4. What I built

Task part	What I did
Agent team	Planner, Researcher, Writer and Critic as LangGraph nodes with shared state (goal, questions, findings, draft, issues); conditional edge Critic -> Writer for one revision
Tools via MCP	Researcher calls describe_data, query_customers and count_by on the Day 16 server (300 synthetic customers)
A2A	Researcher wrapped as an A2A server (official a2a-sdk 1.3): Agent Card + JSON-RPC endpoint at 127.0.0.1:9999; the Planner discovers it and delegates each question as an A2A message
Comparison	Same 5 research tasks run by a single agent (Day 15 harness) and by the team; an LLM judge scores 1-5 against **ground truth computed with SQL**
AI Agent Office	Web app (Starlette + live event stream) that shows the robots working, the SQL they run and the messages between them; opened through a Colab link and a public Cloudflare tunnel

5. Results

Demo run ("Which subscription plan should we focus on to reduce churn, and why?"): 3 sub-questions, 9 LLM calls, 9,767 tokens, 25 s, Critic approved with 0 revisions.

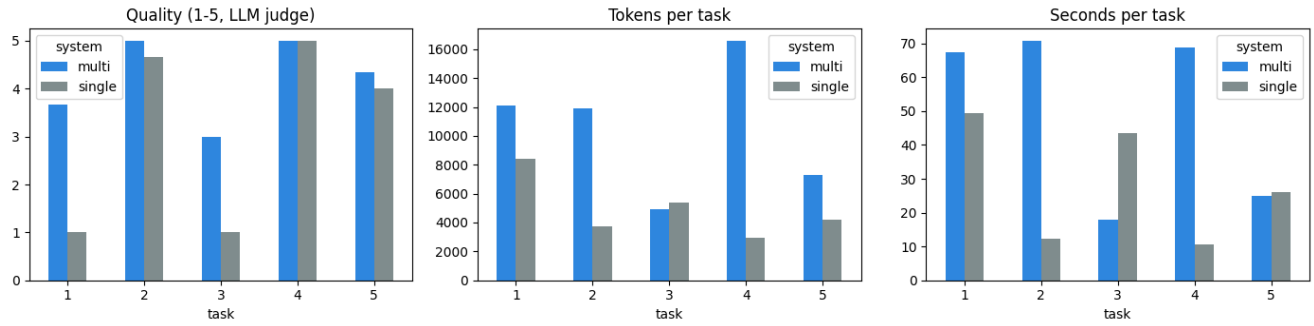
Agent	LLM calls	Tokens	Seconds
Planner	1	395	1.5
Researcher	6	6,844	18.4
Writer	1	1,115	2.4
Critic	1	1,413	2.5

The Researcher uses about 70% of all tokens: it is the only agent that loops over tools.

A2A: the Agent Card was served and read correctly, and the Planner delegated all three questions of "Compare Colombo and Kandy" over A2A (JSON-RPC). The transport worked; the answers had problems (section 6).

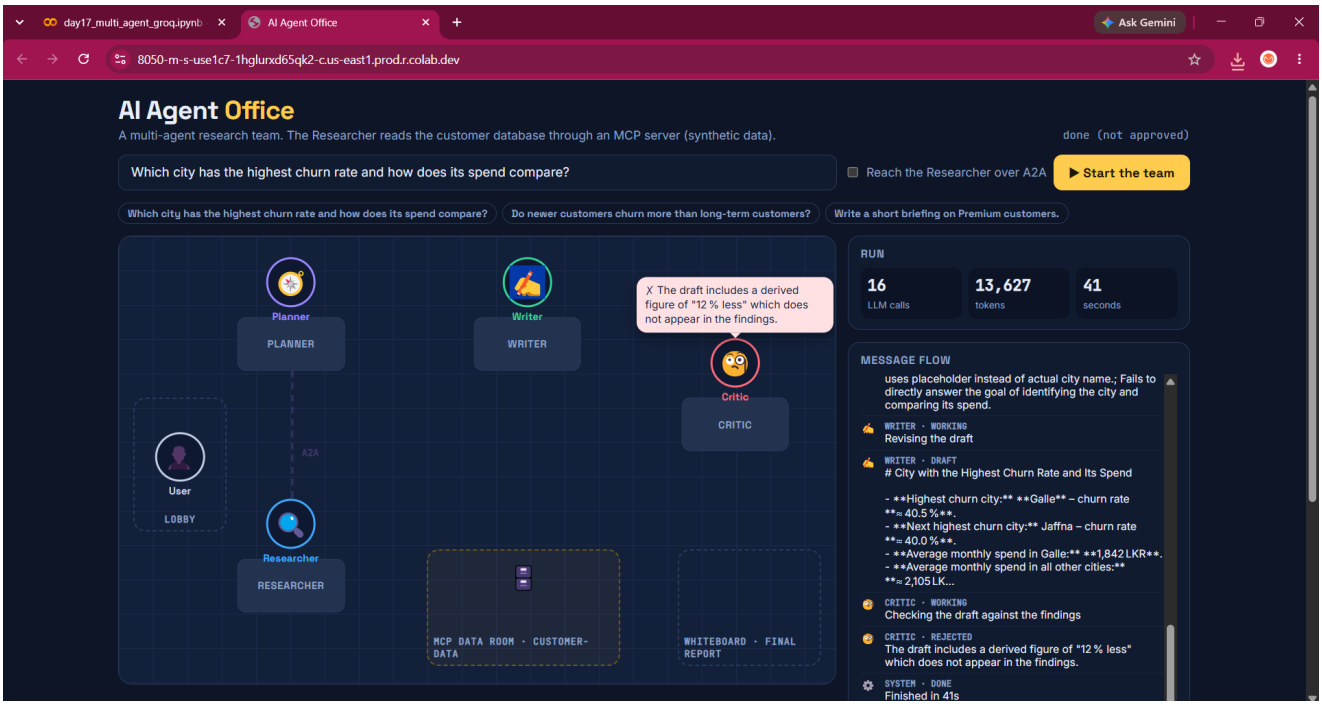
Single agent vs multi-agent (5 tasks, averages):

System	Quality	Correct	Complete	Clear	Tokens	Seconds	LLM calls
Single agent	3.13	2.6	3.4	3.4	4,940	28.3	5.0
Multi-agent	4.20	3.2	4.6	4.8	10,562	50.0	12.6



Quality, tokens and time per task (blue = multi-agent, grey = single agent)

The team scored higher on 4 of 5 tasks and tied on task 4 (and the single agent gave no usable answer on tasks 1 and 3), but it used **2.1x the tokens** and **1.8x the time**. The biggest gains were in *complete* and *clear* (Writer and Critic); *correct* improved only a little (2.6 to 3.2).



AI Agent Office: the Critic rejects a draft because a figure is not in the findings

Office run ("Which city has the highest churn rate and how does its spend compare?"): 16 LLM calls, 13,627 tokens, 41 s. The Critic rejected the draft twice, so the run ended **not approved**.

6. Fact check against the database

I recomputed the true numbers with SQL on the same synthetic database and compared them with what the team reported:

Run	Team said	Database truth	Verdict
Demo: churn by plan	Basic 12%, Standard 8%, Premium 5%	Basic 31.6%, Standard 35.8%, Premium 15.4%	Wrong (invented)
A2A: Colombo vs Kandy spend	12,345 vs 9,876 LKR	2,775 vs 3,439 LKR	Wrong (invented)
A2A: Colombo vs Kandy churn	12.5% vs 8.3%	24.4% vs 28.6%	Wrong (invented)
A2A: customer counts	"data not provided"	Colombo 41, Kandy 49	Missing (step budget)
Office: highest churn city	Galle 40.5%, next Jaffna 40.0%	Galle 40.5%, Jaffna 40.0%	Correct
Office: spend comparison	Galle 1,842 vs others 2,105 LKR	Galle 3,145 vs others 3,113 LKR	Wrong

Main finding: when the Researcher answered **without calling a tool**, the model made up plausible-looking numbers (e.g. 12,345 and 9,876). The Critic only compares the draft with the findings, so it **approved** two reports built on invented numbers. When the Researcher did call the SQL tool, the numbers were right.

7. Failure modes I saw

	Single agent	Multi-agent
Typical failure	Ran out of steps or left placeholders (tasks 1 and 3: no usable answer)	Researcher answered from memory instead of the database; errors passed downstream to Writer and Critic
Critic	-	Too lenient on invented findings; strict on derived numbers (rejected "12% less")
Cost	Lower	About 2x tokens and time; more calls hit the rate limit (30 s waits)
Debugging	One trace	Several agents; needs the message flow / traces to see where it went wrong

8. When is multi-agent worth it?

- When the task has clearly different parts (plan, search, write, review) and quality matters more than cost and speed.
- When parts need different tools or permissions (only the Researcher touches data).
- When agents are owned by different teams or services: that is where A2A helps.
- For short, simple questions a single agent is cheaper, faster and almost as good (task 4: both 5/5).

9. Next improvements

- Force grounding: reject a Researcher answer that used no tool call, and ask again.
- Give the Critic the tool results (SQL rows), not just the summarised findings, so it can catch invented numbers.
- Raise the Researcher step budget from 6 to 8 (the count question stopped at max_steps).
- Keep the SQL ground-truth check in the evaluation: the LLM judge alone would miss these errors.

10. Conclusion

The four-agent team, MCP tools, A2A delegation and the live office all worked end to end. The team wrote more complete and clearer reports than one agent (quality 4.20 vs 3.13) at about twice the cost. The most important lesson: more agents do not make the facts more correct. Every agent must be grounded in tool results, and the reviewer must check against the source data, otherwise a confident team can pass on invented numbers.